Herbert Wertheim College of Engineering
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EIN 4360 Facilities Planning and Material Handling

Chapter 7: Warehouse Operations and Layout

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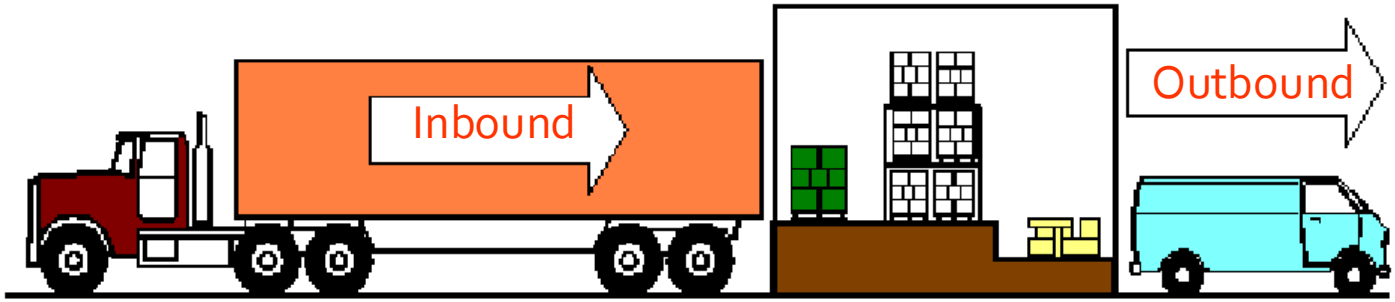
POWERING THE NEW ENGINEER TO TRANSFORM THE FUTURE

Summary

- Objective, mission, and functions of warehouses and warehouse operations
- Types of warehouses
- Traditional vs. *Crossdocking* operations
- Receiving and shipping operations and principles
- Storage models (dedicated vs. randomized)
- *Stock keeping units (SKUs), honeycombing, loss of cube utilization*
- Storage Layout Planning Principles
- Types of storage systems and their advantages and disadvantages
- Warehouse Layout Model for dedicated Storage

Objectives

A **warehouse** is a buffer between inbound orders arriving from suppliers and outbound orders delivered to customers



Missions of a Warehouse

- Critical role of warehouse in **supporting** a company's **supply chain's success**.
- The **mission of a warehouse** is to **ship products to the next step in the supply chain.**
- All warehousing opportunities, including order picking, cross-docking, productivity space utilization, and value-added services, allow the warehouse to process and ship orders more effectively.

Missions of a Warehouse

Warehousing opportunities for improvement are:

1. Improving **order picking** operations
   - Traditionally where a company spends the most time
2. Utilizing **cross-docking**
   - Product comes in and immediately out. (no processing or storage)
3. Increasing **productivity**
   - Labor, equipment, and space and their combinations (e.g., the entire system)
4. Utilizing **space**
   - Rule of thumb is that if a warehouse is more than 80% capacity it is time to add more space or move. This based on the idea that at that capacity it takes longer to put away or find a part.
5. Increasing **value-added services**
   - More than just picking and shipping locations. Now including pre-labeling, sorting, etc.

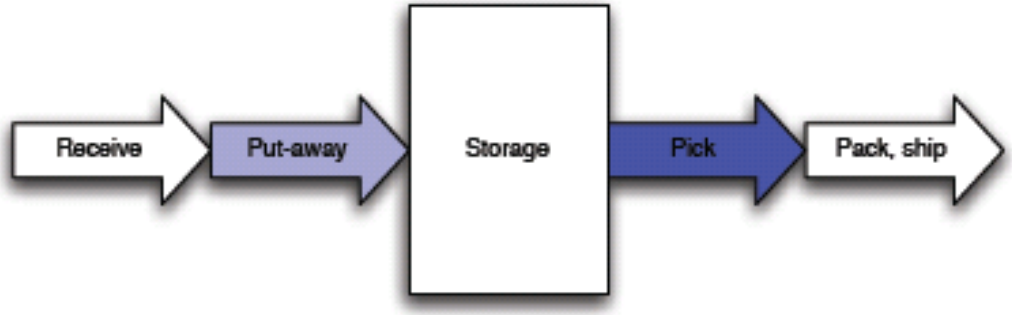
Types of Warehouses

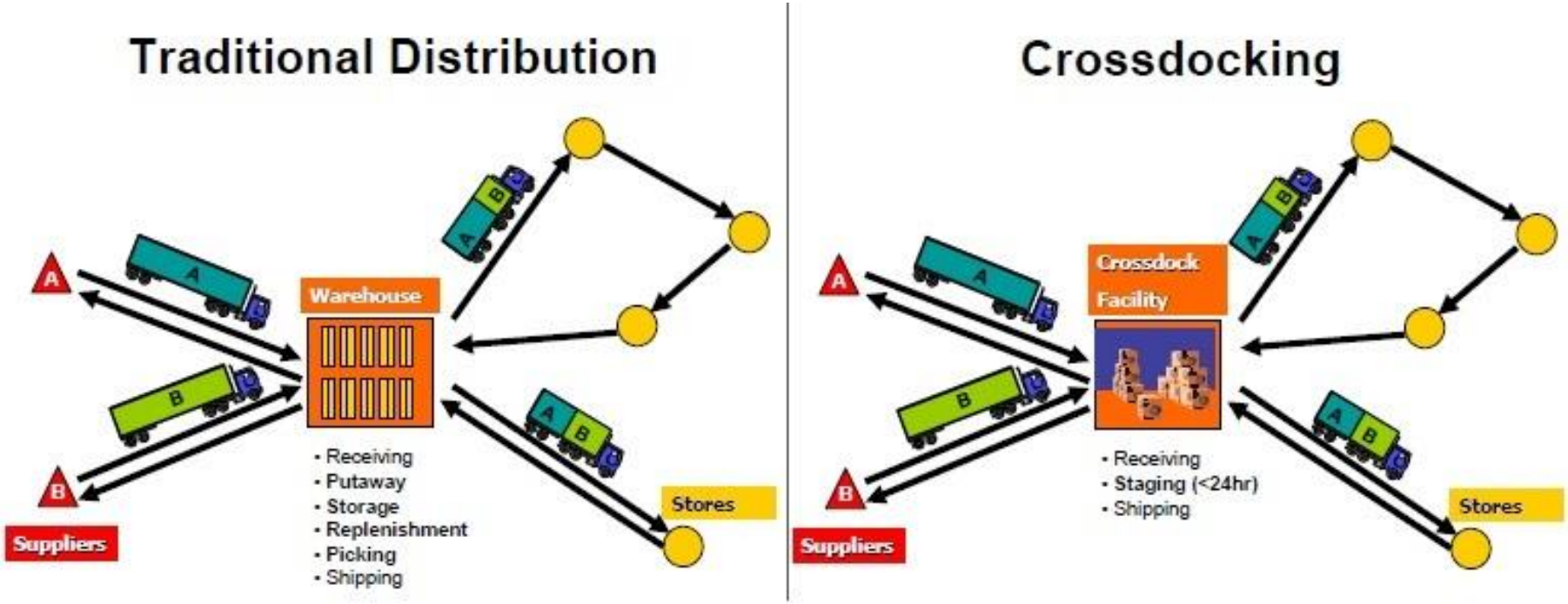
- Retail Distribution Center
  - Ships large orders on regular basis
  - Cross dock – Wal-Mart
- Service Parts Distribution Center
  - Requires storage of many slow-moving parts at irregular intervals
  - Large plane parts
- Catalog Fulfillment or E-commerce
  - Receives many small orders of many items
  - Amazon
- 3PL (Third Party Logistics) Warehouse
  - Companies may outsource warehousing
  - Fulfilment by Amazon

High tech Warehouse (Example):

Operations of a Warehouse

- Inbound Processes
  - Receiving
  - Put-away
- Outbound
  - Order-Picking
  - Packing, Shipping



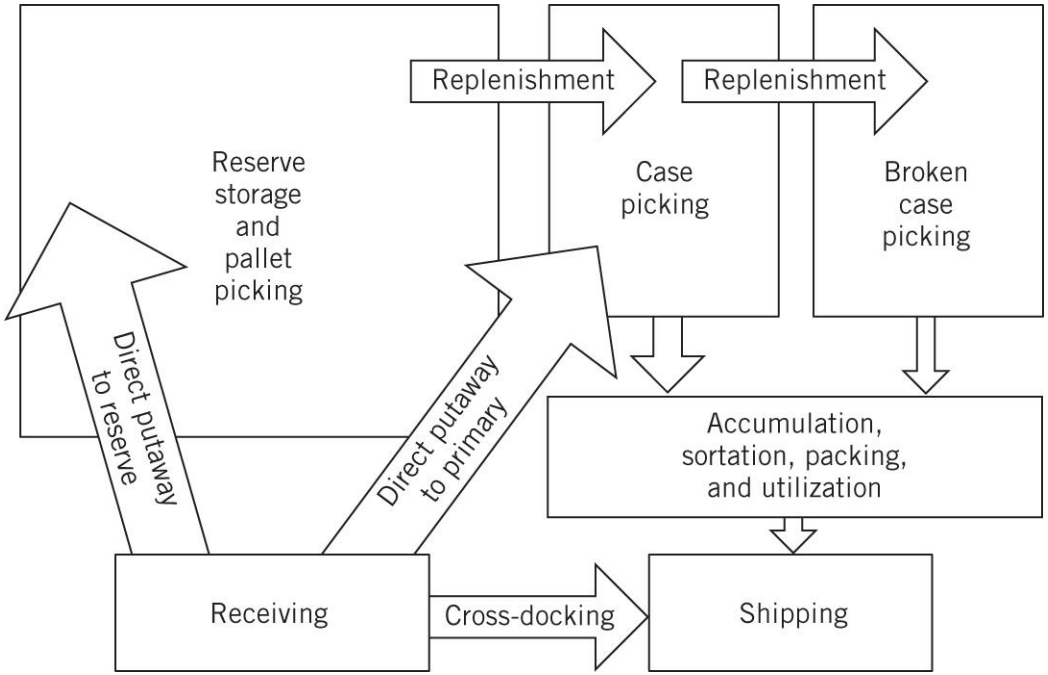


Source: http://people.sabanciuniv.edu/ertekg/papers/2005

Functions in the Warehouse

There are many activities that occur as part of the process of getting material into and out of the warehouse

1. Receiving
2. Inspection and Quality control
3. Repackaging
4. Put-away
5. Storage
6. Order picking
7. Postponement
8. Sortation of batch
9. Packing and Shipping
10. Cross-docking
11. Replenishing



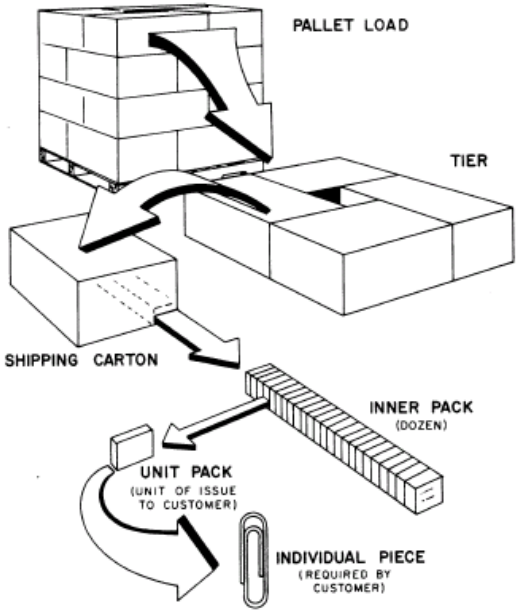
Functions in the Warehouse

What is the difference between a warehouse and a distribution center?

- Warehouse: Storage plays a large role
- Distribution Center: Little or no storage

Stock Keeping Unit (SKU) – The smallest physical unit of a product tracked by an organization

In general, the smaller the handling unit, the larger the handling costs



Functions in the Warehouse

How do we measure the performance of warehouse based on its functions and compare it with other warehouses?

- Order accuracy
- Order fill rate
- On-time delivery
- Line items picked per hour
- Percent breakdown of labor hours expended on receiving, put-away, picking, replenishment, etc.

- Important to consider a set of metrics and not just one in isolation

Receiving and Shipping Operations

What activities are required to **receive** the goods?

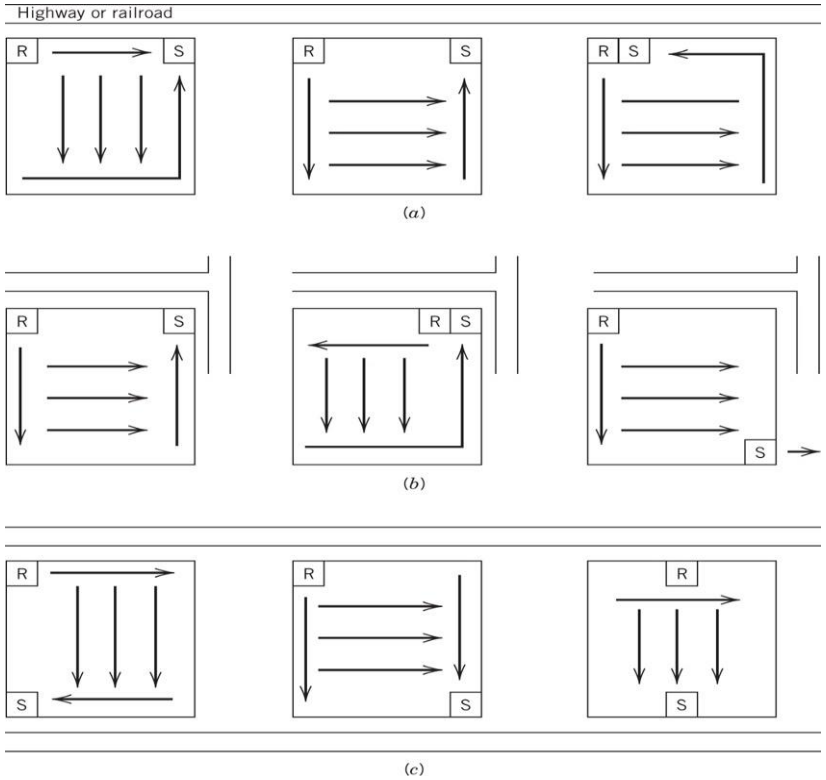
- Inbound trucker notifies warehouse in advance
- Warehouse receiving personnel verifies order
- Trucker arrives and is assigned to a door (dock)
- Vehicle is secured and cargo is inspected
- Unitized and Floor-Loaded or Loose cargo is unloaded
- All unloaded material is staged and counted
- Proper disposal of damaged goods is made
- Load is staged for put-away

See how many times the product is touched…can you minimize it?

The natural sequence for the flow of materials is:
>vendor > receiving > storage > production > warehousing > shipping > customer

Receiving and Shipping Operations

The location of receiving and shipping depends on access to transportation facilities



Desired Attributes of Receiving:

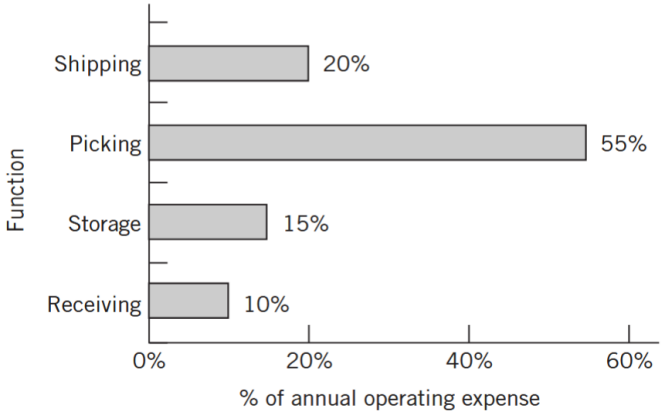
- Directed flow paths (**avoid double handling**)
- Continuous flow of effort (try to schedule inbound shipments)
- Concentrated area that minimizes material handling
- Efficient material handling (try to influence unit load consideration)
- Provide **safe** operations
- **Minimize damage**
- Good housekeeping

Receiving Principles

The following principles serve as guidelines for streamlining receiving operations to simplify the flow of material through the receiving process and to ensure the minimum work is required.

1. **Don't receive:** (e.g., Apple's drop shipping, or for heavy/bulky items)
2. **Pre-receive:** Process incoming shipments before they arrive rather than during staging
3. **Cross-dock material**
4. **Put away**,….

Receiving accounts for approximately 10% of warehouse operating expenses.



Shipping Principles

Many of the best-practice receiving principles also apply in reverse to shipping, including direct loading, advanced shipping notice preparation and staging in racks.

1. Select cost and space effective handling units
2. Minimize product damage
3. Eliminate shipping staging
4. Use storage racks
5. Route on-site drivers through the site and minimize paperwork and time.
6. Use small-parcel shipping

Shipping accounts for about 20% of warehouse operating expenses.

Receiving and Shipping

- Other Considerations:
  - Smooth interface between vendor's and receivers' (shipper and customer's) information system
  - Returnable containers
  - Returned goods
  - Returning carriers, i.e. backhaul (maximize utilization of shipper owned carriers)
- Coordination between Receiving and Shipping
  - Common space, equipment and personnel might be used
  - Return of base pallets for manufacturing
  - Possible centralization of the function (ship in the morning, receive in the afternoon)

In-class Activity

Match the warehouse function/activity with its respective match:

1. **Inspection & Quality control**
2. **Repackaging**
3. **Put-away**
4. **Storage**
5. **Order picking**
6. **Postponement**
7. **Sortation**
8. **Cross-docking**
9. **Replenishing**

- When a product is received in bulk from a supplier and must be packaged singly.
- Refill primary picking shelves from bulk storage.
- Select items from shelves for customer orders.
- Transfer goods directly from inbound to outbound trucks.
- Moving goods from receiving to assigned storage racks.
- Check items for damage or defects when supplier quality is inconsistent.
- The physical containment of the items, depends on size & quantity.
- Add final labels, like prices, right before shipping.
- Separate mixed-picked items into individual orders.

In-class Activity

Answer

1. **Inspection & Quality control:** Check items for damage or defects when supplier quality is inconsistent.
2. **Repackaging:** When a product is received in bulk from a supplier and must be packaged singly.
3. **Put-away:** Moving goods from receiving to assigned storage racks.
4. **Storage:** The physical containment of the items, depends on size & quantity.
5. **Order picking:** Select items from shelves for customer orders.
6. **Postponement:** Add final labels, like prices, right before shipping.
7. **Sortation:** Separate mixed-picked items into individual orders.
8. **Cross-docking:** Transfer goods directly from inbound to outbound trucks.
9. **Replenishing:** Refill primary picking shelves from bulk storage.

Facility Requirements for Receiving & Shipping

- Sufficient area to stage orders
  - Waiting
  - Maneuvering
  - Unloading
- Docks
  - Must be able to adjust to facilitate carrier loading

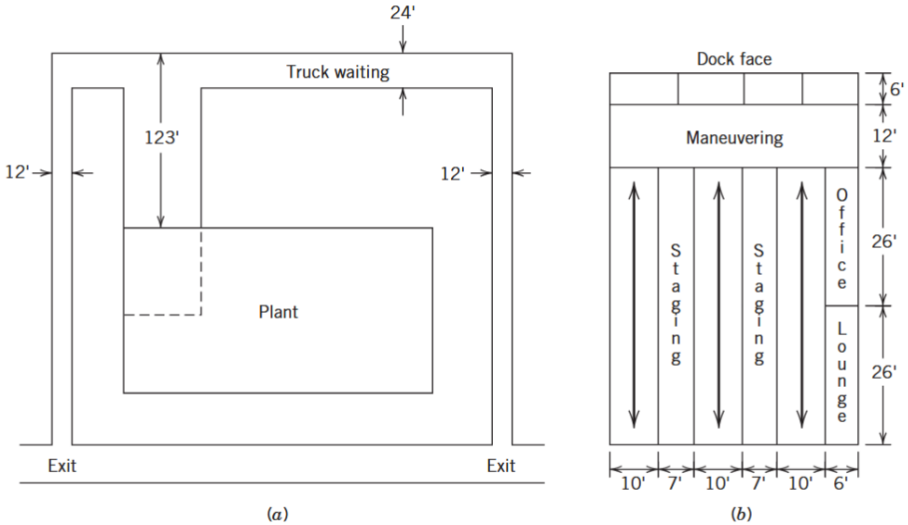


Figure 7.9 (*a*) Site and (*b*) dock area for Example 7.1.

Receiving Space Planning

- What is to be received and shipped
  - Receiving and shipping analysis chart
- Number and types of docks
  - Estimate with simulation, queuing analysis
  - Configuration
    - Carrier approaches, traffic flow and waiting area
    - Apron depth, bay width
- Space requirements for shipping and receiving area
  - Personnel Convenience, Offices, Material Handling Equipment Maintenance, Trash Disposal, Pallet and Packing Material Storage, Trucker's Lounge, Staging Area, Equipment Maneuvering

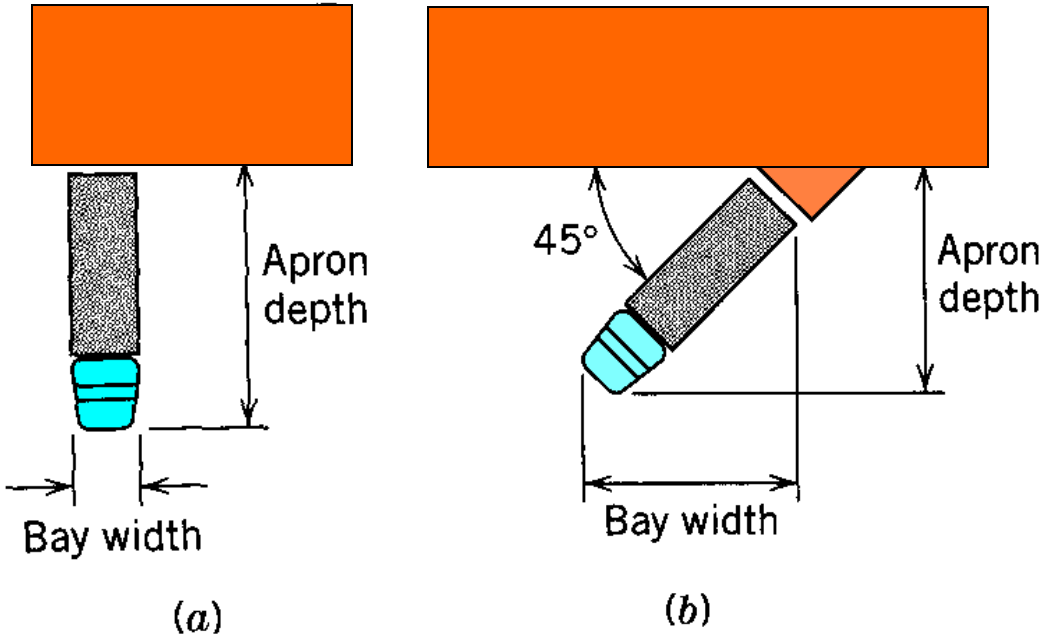
Company _____ Date _____ Raw Materials _____ Finished Goods _____
Prepared by _____ Sheet _____ of _____ Plant Supplies _____

1	2	3	4	5	6	7	8	9	10	11
	UNIT LOADS				Size of Shipment (unit loads)	Frequency of Shipment	TRANSPORTATION		MATERIAL HANDLING	
Description	Type	Capacity	Size	Weight			Mode	Specifications	Method	Time

Figure 7.4 Receiving and shipping analysis chart.

Number & Types of Docks

Docks are places for loading or unloading material. There are two general types: 90° docks and finger docks (Figure 7.8). Tables 7.1 and 7.2 summarize space requirements for both types.



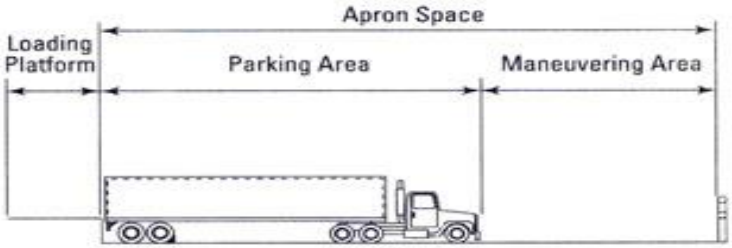


Fig. 6

Determining Number of Docks

S_f safety factor
T_{day} average number of trucks per day
H_{truck} average hours to load/unload a truck
H_{day} average hours available per day

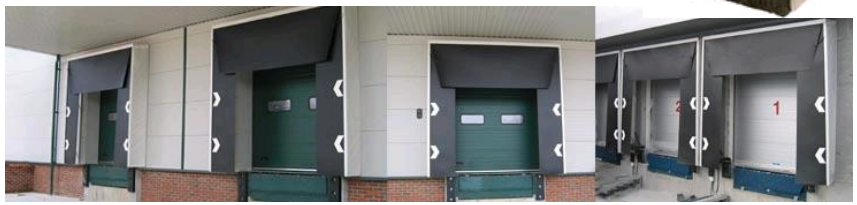
$$N_{dock} = \frac{S_f T_{day} H_{truck}}{H_{day}}$$

The *safety factor* depends on arrival rates, load/unload times, size of marshaling area, and desired maximum waiting time in the marshaling area. It basically asks how conservative do we want to be?

How do we get material into and out of the trucks?

Ramps and Lifts

Seals



Bumper Pads

STORAGE OPTIONS

Storage Operations

The objective of storage and warehousing functions is either

- To **maximize resource utilization** while satisfying customer requirements
- **Maximize customer satisfaction** subject to a resource constraint

In designing storage and warehousing systems, it is desirable to maximize

- Space utilization
- Equipment utilization
- Labor utilization
- Material accessibility
- Material protection

Storage Space Planning

There are two major material storage philosophies:

- **Dedicated** Storage: fixed location storage
  - Each SKU is assigned to a specific storage location or set of locations.
  - Storage locations can be arbitrarily determined, such as part number sequence, or they can be determined based on the SKU's activity level and inventory level.
  - There must be sufficient space to store the maximum amount of the SKU that will be on hand.
- **Random** or **floating** location storage
  - An individual SKU can be stored in any available storage location.
  - An inbound load is assigned to the closest available storage location.
  - Retrievals are first-in, first-out (FIFO).
  - Maximizes space utilization

Storage Space Planning

- The amount of space planned for an SKU (Stock Keeping Unit) is directly related to the method of assigning space.

- If **fixed** location storage is used, then a given SKU must be assigned sufficient space to store the **maximum amount that will ever be on hand.**
  - Maximizes throughput because we know exactly where things are
  - SKU with highest ranking is put closest to the I/O point

- For **random** location storage, the quantity of items on hand at any time will be the **average amount of each SKU**.
  - If a large safety stock of SKUs is maintained, then the quantity of items on hand at any time will be the safety stock plus one-half the <u>maximum</u> replenishment quantity of each individual SKU
  - Replenishment Quantity: The average of what cycles in and out

Storage Space Planning

- To maximize throughput when using **dedicated storage**, SKUs should be assigned to storage locations based on the ratio of their activity (number of storages/retrieval per unit time) to the number of openings or slots assigned to the SKU.
- The SKU having the highest ranking is assigned to the preferred openings and so on, with the lowest-ranking SKU being assigned to the least-preferred openings.
- Because "fast movers" are up front and "slow movers" are in back, throughput is maximized.

Storage Space Planning

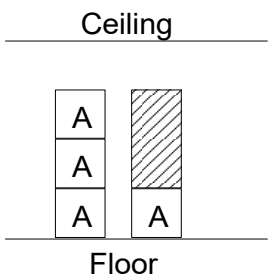
- Despite the greater throughput of dedicated storage, it is not used as often as it should be.
- One reason is that it requires more information to plan the system for maximum efficiency.
- When conditions change significantly, items must be relocated to achieve the benefits of dedicated storage.
- Hence, randomized storage is more appropriately used under highly seasonal and dynamic conditions.

Class-based Storage Space Planning

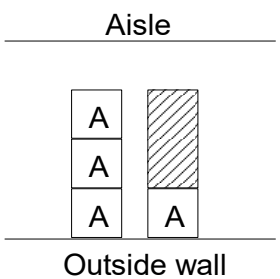
- When many SKUs exist, dedicated storage based on each SKU may not be practical.
- Instead, SKUs can be assigned to classes based on their activity-to-space ratios.
- **Class-based dedicated storage**, with randomized storage within the class, can yield the throughput benefits of dedicated SKU storage and the space benefits of randomized SKU storage.
- Depending on the activity-to-space ratios, three to five classes might be defined.

Honeycombing

- **Honeycombing**
  - Wasted space that results because a partial row or stack cannot be utilized because adding materials would result in blocked storage.



Vertical Honeycombing

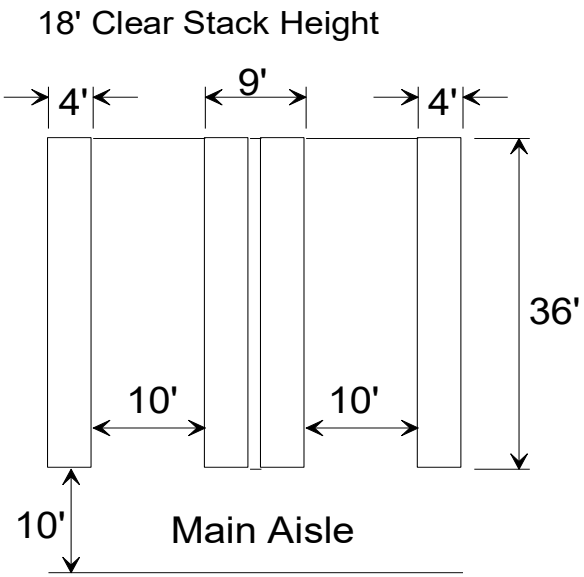


Horizontal Honeycombing

- **Space standard**
  - **The amount of space each unit load needs**, including the extra space required for **aisles, access, and any empty or unused space (honeycombing)** in the storage area.

Loss of Cube Utilization

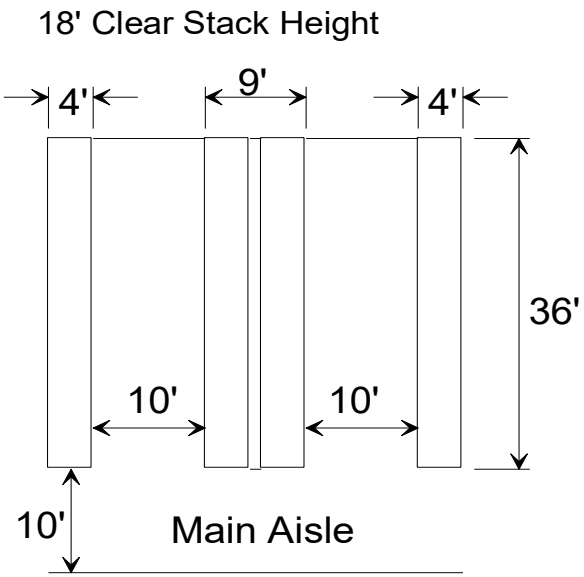
- **Aisle Example**



Total Cube =
Aisle Cube =
Percentage of total volume allocated to aisles =

Loss of Cube Utilization

- **Aisle Example**



4+10+9+10+4= 37

36+10=46

Total Cube:

46*37*18=30636

Aisle Cube:

10*36*18*2+10*37*18=19620

Total Cube = 30,636 cubic feet

Aisle Cube = 19,620 cubic feet

Percentage of total volume allocated to aisles = 64%

Storage Layout Planning

Objectives of a warehouse layout:

- ✓ To use space efficiently
- ✓ To allow for the most efficient material handling
- ✓ Provide most economic storage
- ✓ Provide maximum flexibility
- ✓ To make the warehouse a model of good housekeeping

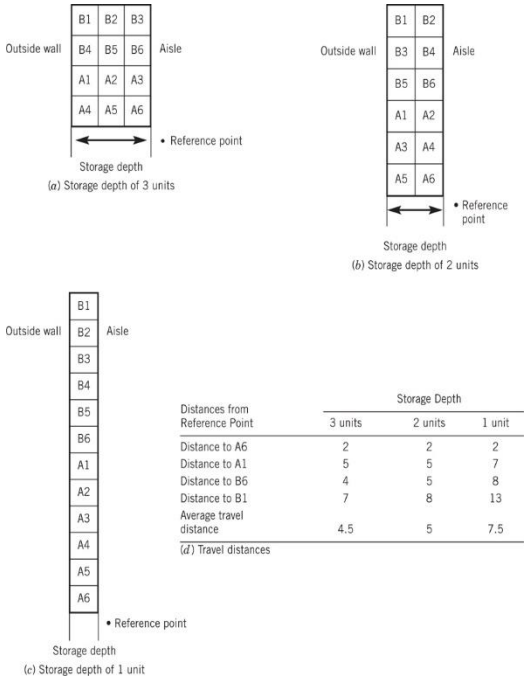
Storage Layout Planning

To accomplish objectives several storage area principles must be integrated:

Popularity:

- **Pareto's Law** applies to the popularity of materials stored: typically, 85% of the turnover will be a result of 15% of the materials stored.

- To maximize throughput, the most popular 15% of material should be stored such that travel distance is minimized



Distances from Reference Point	Storage Depth 3 units	2 units	1 unit
Distance to A6	2	2	2
Distance to A1	5	5	7
Distance to B6	4	5	8
Distance to B1	7	8	13
Average travel distance	4.5	5	7.5

(d) Travel distances

Storage Layout Planning Principles

- **Similarity**
  - Items that are received and shipped together should be stored together.
    - Reduce distance pickers need to travel.

- **Size**
  - Provide a variety of storage location sizes to accommodate a variety of products.
  - Utilize adjustable racks or shelves if product and unit load sizes are uncertain or change periodically.

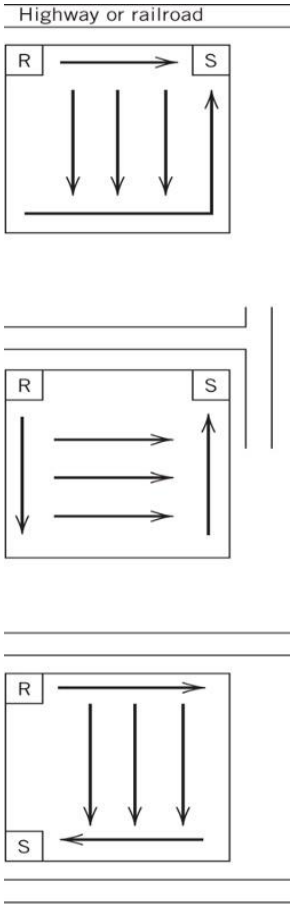
Storage Layout Planning Principles

- **Characteristics**
  - **Perishable material**: required a controlled environment and shelf life of materials should be considered.
  - **Oddly shaped and crushable items**: open storage or adjusted storage and methods should be provided.
  - **Hazardous material**: paint, flammable chemicals,... require separate storage, safety codes should be checked for flammable and explosive materials. Exposure of employees to acids,... must be minimized.
  - **Security**: some items of high unit value and small size are often targets for pilferage and theft
  - **Compatibility**: some materials do not require special storage, but become easily contaminated if allowed to come in contact with certain other materials.

Storage Layout Planning Principles

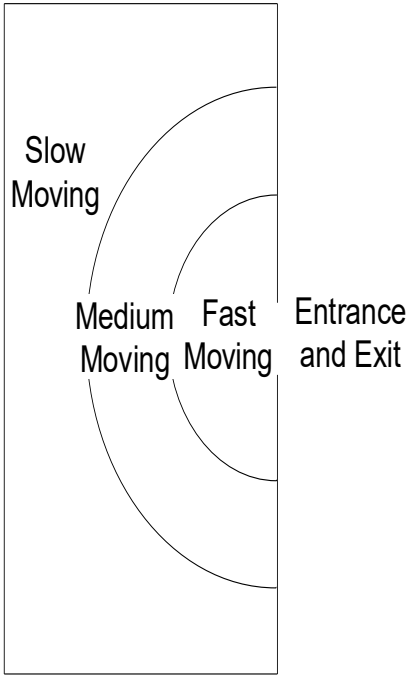
- **Space Utilization**
  - **Conservation of Space**
    - Maximizing concentration
    - Maximizing cube utilization
    - Minimizing honeycombing
  - **Limitations of Space**
    - Clear height
    - Stacking height
    - Floor loading
    - Columns
- **Accessibility of Stored Materials**
  - **Each storage face has aisle access**
  - **Majority of items stored along the long axis of the area**
  - **Aisles should not be placed along walls without doors**
  - **Avoid locked stock by using a two-bin system**
    - When first bin is empty, it is returned to the supplying process to refill. The second bin supplies parts while the first one is being filled

Storage Layout Planning



If materials enter and leave the warehouse

- From the **same** point ➔ positioned as close to this point as possible.
- From **different** points received and shipped in the same quantity ➔ positioned along the most direct route between the entrance and departure point.
- From **different** points and are received and shipped in different quantity:
  - Smallest r/s ratio ➔ positioned close to the shipping point
  - Largest r/s ratio ➔ positioned close to the receiving point



In-class Activity

A manufacturing facility is open **10 hours per day**, during which time an average of **25 trucks arrive** daily for receiving and an average of **22 trucks depart** daily for shipping. It takes an average of **45 minutes to unload** a truck and **90 minutes to load** a truck. The management has established a **safety factor of 110%.** How many dock doors are need for this facility?

$$N_{dock} = \frac{S_f T_{day} H_{truck}}{H_{day}}$$

In-class Activity

Answer: If resources are not shared b/w R & S

$$N_{rec} = \frac{1.1 * 25 * 0.75}{10} = 2.06 \rightarrow 3$$

$$N_{shp} = \frac{1.1 * 22 * 1.5}{10} = 3.63 \rightarrow 4$$

$$N_{docks} = 3 + 4 = 7$$

In-class Activity

Alt. Answer: If resources are shared b/w R & S

$$N_{dock} = \frac{1.1 * (25 * 0.75 + 22 * 1.5)}{10} = 5.69 \rightarrow 6$$

STORAGE TECHNOLOGY

Rack Storage Systems

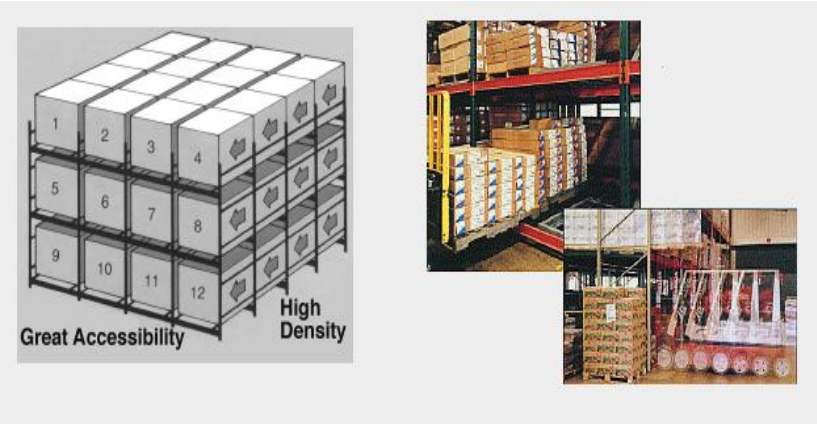
- Primary method of storage
- Utilizes vertical space
- Highly cost efficient
- Types:
  - Walk through
  - Push Back
  - Very Narrow Aisle
  - Drive-in
  - Mobile
  - Double Deep
  - Rollout Shelf
  - Reel
  - Cantilever
  - Gravity

Walk Through Rack

- Provides easy accessibility to the adjacent aisles
- Picking time is greatly reduced
- Allows more lighting into aisles
- Higher utilization of vertical space

Push Back Racks

- Pallet loads are literally pushed back into the rack
- When pallet is retrieved, the deeper pallet load automatically advances to the aisle
- High density and high accessibility
- Used to store large number of pallets for a long time and then remove and ship them as seasonal products
- Used mostly in combination with drive-in racks

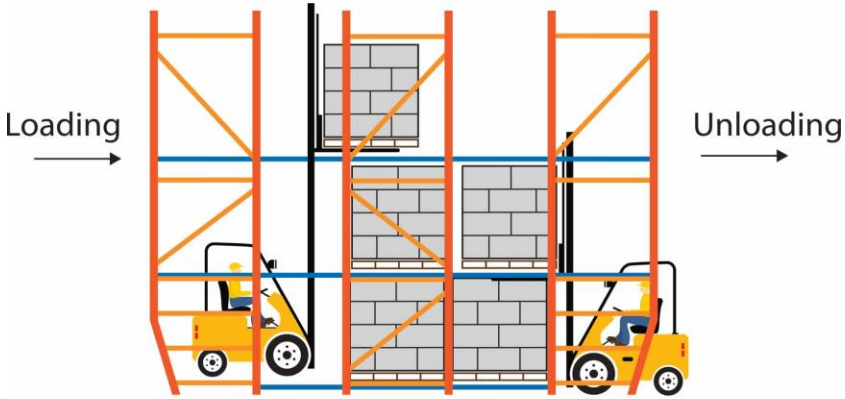


Very Narrow Aisle Racks

- Permits aisle widths less than 5 feet
- Provide significant floor space reduction
- Used at heights of 40 feet to 50 feet
- Pallets are accessed with turret or side loader lift trucks (specialized equipment)
- Sometimes racks are designed with top and bottom monorails

Drive-In Racks

- Used for bulky and light weight parts
- Parts are mainly in cartons
- Fork lift is the material handling device used
- Parts are typically stacked one over the other in large numbers



Drive-Through Rack

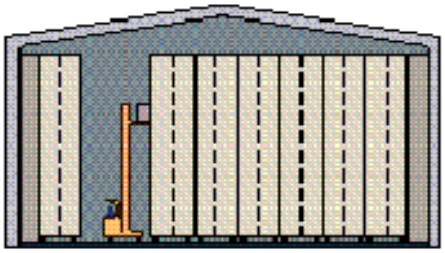
Gravity Flow Racks

- Used for parts in cartons
- Parts are supported on rollers in the rack system
- Material transfer is done with ease
- Easy to move parts along the same row as less effort is required due to the presence of rollers

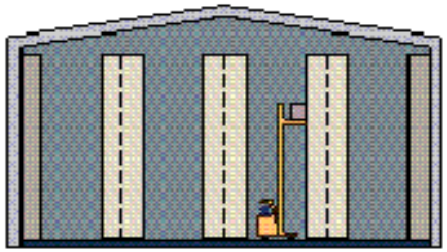
https://www.youtube.com/watch?v=K3tPx8oPlxE

Mobile Racks

- High density storage systems
- 100% utilization of pallet positions
- Storage carriages are driven by electric motors
- Used mainly in cold or ultra-cold warehouses where aisle space is at a premium
- Used when speed of product movement is not a major concern



Single-aisle, mobile-rack storage



Conventional multiple-aisle rack storage

Double Deep Racks

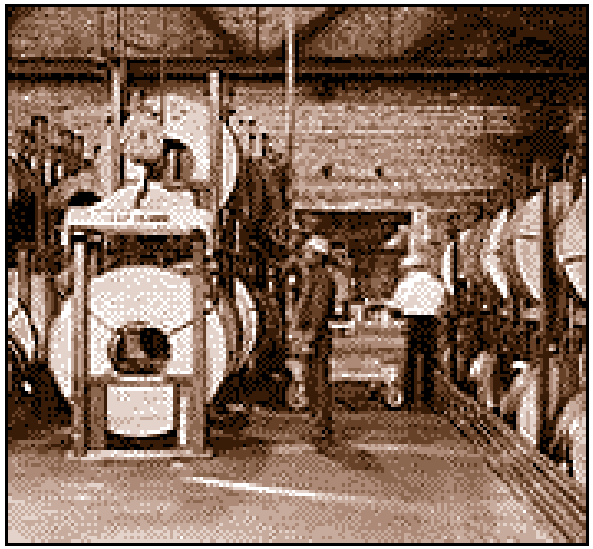
- It is like placing two rows of racks together
- Uses a deep reach lift truck for storage and retrieval
- Increases floor space usage to about 60% to 65%
- Used for high throughput operations
- Low initial cost
- High productivity
- Less equipment damage

Rollout Shelf Racks

- Specialized storage
- Mainly for delicate and costly parts
- Used for dies, measurement tools

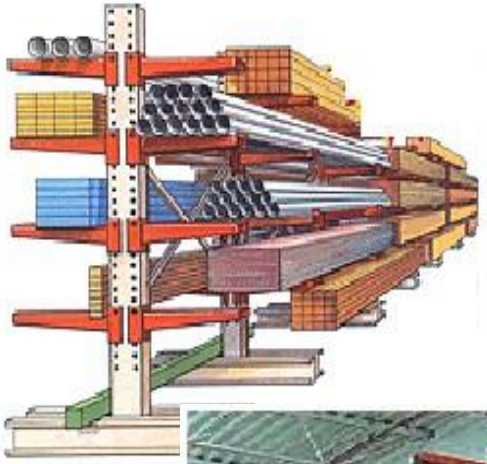
Reel Racks

- Special type of storage
- Mainly for cables and wires in reels
- Easy to retrieve varying amounts

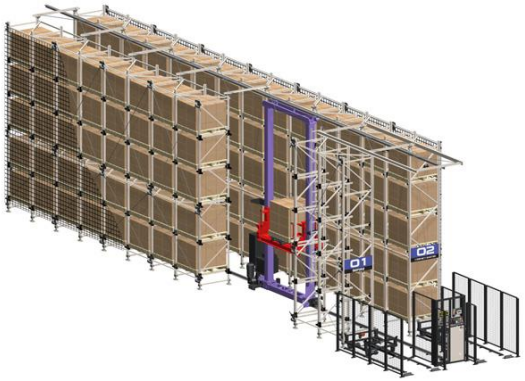


Cantilever Racks

- Used to store long pipes and wood
- Parts are stored using the fork lifts
- Highly economical
- Can be used till heights of 22 feet



Automated Storage and Retrieval Systems: AS/RS Machines



- Improved Space Utilization
- 75 % of rack space can be recovered
- High saving in pickers time
- Allows quick access to goods via a system of shelves operating on a shuttle mechanism
- It brings the exact pick/store location to the operator
- Increases retrieval productivity by more than 2.5 times
- Access area of the system is ergonomically designed to present stored items at an ideal height for picking
- High employee safety and improved throughput

https://www.youtube.com/watch?v=Ox05Bks2Q3s

Carousel Storage

https://www.youtube.com/watch?v=uueNp0eUf70

- Advantages:
  - High pick rates
  - Motorized, computer controlled and independent rotating aisles of shelving
  - Parts are carried to the picker rather than the picker going to the parts like in the case of racks
  - Mostly set up in pods of 2 or 3
  - Pick rates vary from 80-200 picks/person hour

Carousel Storage

DISADVANTAGES

- Adding more people cannot significantly increase pick rate
- Only one picker can operate at a given time
- Thus reduces the ability of the warehouse to respond to surges in demand

STORAGE PLANNING MODELS

Next: Two Storage Planning Models

- Dedicated Storage Planning
  - Warehouse Layout Model (10.3.2)
- Randomized Storage Planning
  - Automated Storage and Retrieval System (AS/RS)

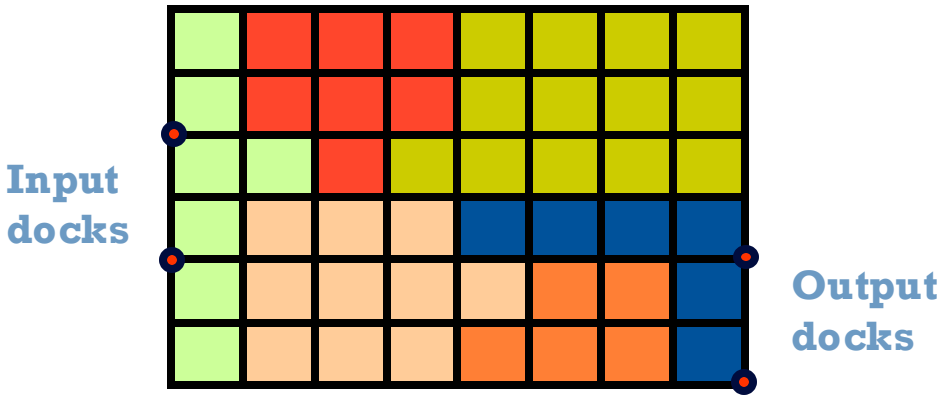
Recall: Types of Storage

(1) **Randomized or Floating Storage**: Used when a product unit (SKU - stock-keeping unit) can be stored in any available storage location.

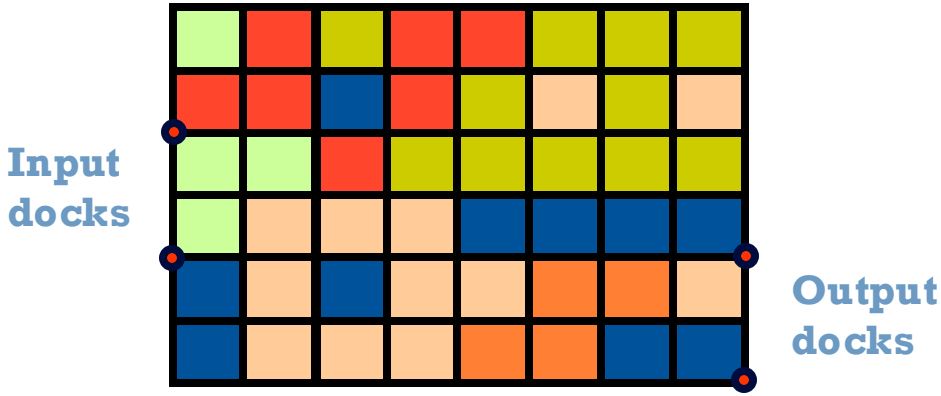
(2) **Dedicated Storage**: Used when a unit of product (SKU) is assigned to a specific storage location or set of locations.

- **Variations of Dedicated Storage**: Storage according to part number sequence, and storage according to level of activity (*storage/retrievals per unit time*) and inventory level.

Types of Storage



Dedicated Storage



Random Storage

Storage Planning Models

Ex 1: Determining storage space requirements in number of loads

At the ABC manufacturing firm, the average daily withdrawal rate for product A is 20 cartons per day, safety stock is 5 days, order lead time is 10 days, and the order quantity is 45 days. What are the maximum and average quantities of unit loads to be stored?

Reorder point=(safety stock in days)(demand/day)+(lead time in days)(demand/day)

Maximum quantity to be stored=(safety stock)+(order quantity)

Average quantity to be stored=1/2 (order quantity)+(safety stock)

Storage Planning Models

Ex 1: Determining storage space requirements in number of loads

At the ABC manufacturing firm, the average daily withdrawal rate for product A is 20 cartons per day, safety stock is 5 days, order lead time is 10 days, and the order quantity is 45 days. What are the maximum and average quantities of unit loads to be stored?

Reorder point	=	(safety stock in days)(demand/day) + (lead time in days)(demand/day)
	=	(5 days)(20 cartons per day) + (10 days)(20 cartons per day)
	=	300 cartons
Maximum quantity to be stored	=	(safety stock) + (order quantity)
	=	(100 cartons) + (45 days)(20 cartons per day)
	=	1000 cartons
Average quantity to be stored	=	1/2 (order quantity) + (safety stock)
	=	1/2 (900) + (100)
	=	550 cartons

Storage Planning Models

Ex 2: Randomized vs Dedicated

Dedicated storage:
Storage is the sum of the max inventory levels for each product
24+36+8+24+24+24=140

Random storage:
The maximum aggregate 105

	PRODUCTS						
Period	1	2	3	4	5	6	Aggregate
1	24	12	2	12	11	12	73
2	22	9	8	8	10	9	66
3	20	6	6	4	9	6	51
4	18	3	4	24	8	3	60
5	16	36	2	20	7	24	105
6	14	33	8	16	6	21	98
7	12	30	6	12	5	18	83
8	10	27	4	8	4	15	68
9	8	24	2	4	3	12	53
10	6	21	8	24	2	9	70
11	4	18	6	20	1	6	55
12	2	15	4	16	24	3	64
13	24	12	2	12	23	24	97
14	22	9	8	8	22	21	90
15	20	6	6	4	21	13	75
16	13	3	4	24	20	15	84
17	16	36	2	20	19	12	105
18	14	33	8	16	13	9	98
19	12	30	6	12	17	6	83
20	10	27	4	8	16	3	68
21	8	24	2	4	15	24	77
22	6	21	8	24	14	21	94
23	4	18	6	20	13	18	79
24	2	15	4	16	12	15	64

Maximum of aggregate inventory level = 105 pallet loads
Sum of individual maximum inventory levels = 140
Average inventory level = 77.5
Minimum of aggregate inventory level = 51

WAREHOUSE LAYOUT MODEL

Warehouse Layout Model

Dedicated storage

A set of storage locations assigned to a specific product

Decision variable

$x_{jk} = 1$ if product j is assigned to location k, and 0 otherwise

Measure of effectiveness

Minimize total expected distance traveled

Input Data

q = number of storage locations
n = number of products
m = number of input/output points (docks)
S_j = number of storage locations required by product j
T_j = number of trips in/out of storage for product j
p_i = percentage of travel in/out of storage to/from point i
d_{ik} = distance or time required to travel from point i to location k

(Note: We will use **rectilinear** distance between storage **centroids**)

Mathematical Model

$$\text{Min} \quad \sum_{j=1}^{n} \frac{T_j}{S_j} \sum_{k=1}^{q} \left(\sum_{i=1}^{m} p_i \, d_{ik} \right) x_{jk}$$

Equation Breakdown: (We want to minimize this final value)

For each product **j** in **n**:

Compute the **ratio** of the number of trips, $\mathbf{T_j}$, over the required number of storage locations, $\mathbf{S_j}$, for product **j**

For each storage location **k** in **q**:

The traffic density/intensity for that item

If product **j** is assigned to location **k**:

For each I/O point **i** in **m**:

Compute the product of the percentage of travel through that I/O point, $\mathbf{p_i}$, and distance to travel from **i** to **k**, $\mathbf{d_{ik}}$

The weighted average (expected) distance traveled

$$\text{Min} \quad \sum_{j=1}^{n} \frac{T_j}{S_j} \sum_{k=1}^{q} \left(\sum_{i=1}^{m} p_i \, d_{ik} \right) x_{jk}$$

$$\sum_{j=1}^{n} x_{jk} = 1, \; k = 1, \ldots, q$$

Any one location can only store a single item type

$$\sum_{k=1}^{q} x_{jk} = S_j, \; j = 1, \ldots, n$$

The required number of storage locations for each product is met

$$x_{jk} = \{0,1\} \quad for \; j = 1, \ldots, n; \; k = 1, \ldots, q$$

$$Min \quad \sum_{j=1}^{n} \frac{T_j}{S_j} \sum_{k=1}^{q} \left(\sum_{i=1}^{m} p_i \, d_{ik} \right) x_{jk}$$

$$f_k = \sum_{i=1}^{m} p_i \, d_{ik}$$

$$Min \quad \sum_{j=1}^{n} \frac{T_j}{S_j} \sum_{k=1}^{q} f_k \, x_{jk} = \sum_{j=1}^{n} \frac{T_j}{S_j} \left(f_1 \, x_{j1} + f_2 \, x_{j2} + \ldots + f_q \, x_{jq} \right)$$

Solution Procedure

(a) Re-label/Re-order the products (*1, 2, … n*) according to the *nonincreasing* order of their T_j / S_j values:

$$T_1 / S_1 \geq T_2 / S_2 \geq \ldots \geq T_n / S_n.$$

(b) Find the values of f_k (expected distance traveled between location k and the I/O points)

(c) Assign the locations to products according to their f-values from lowest to highest (lowest f-value gets matched with j that has the largest T_j / S_j and so on…)

Example

Warehouse dimensions: 40 ft x 40 ft
Storage location: 10 ft x 10 ft

- One receiving dock at the northeast corner of the layout
- One shipping dock at the midpoint of the west side

Two products A and B:

Trips in/out of storage

Product A: 100 pallets per week ($T_A = 100$)
Product B: 80 pallets per week ($T_B = 80$)

Storage locations required

Product A requires 10 storage bays ($S_A = 10$)
Product B requires 4 storage bays ($S_B = 4$)

Solution

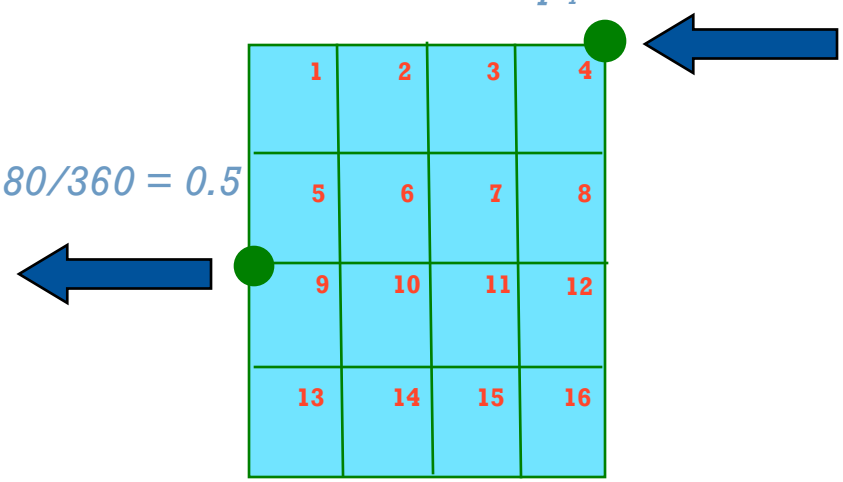
Warehouse dimensions: 40 x 40 ft
Storage location: 10 x 10 ft

Total in and out:

*A: 2*100, B: 2*80*

$p_1 = 180/360 = 0.5$

$p_2 = 180/360 = 0.5$



Compute: $$f_k = \sum_{i=1}^{m} p_i d_{ik}$$

for $k = 1, \ldots, q$

$f_k = p_1 d_{1k} + p_2 d_{2k}$

In this case, what does p tell us about the r/s ratio?

$T_A=100$
$S_A=10$
$T_A/S_A=10$

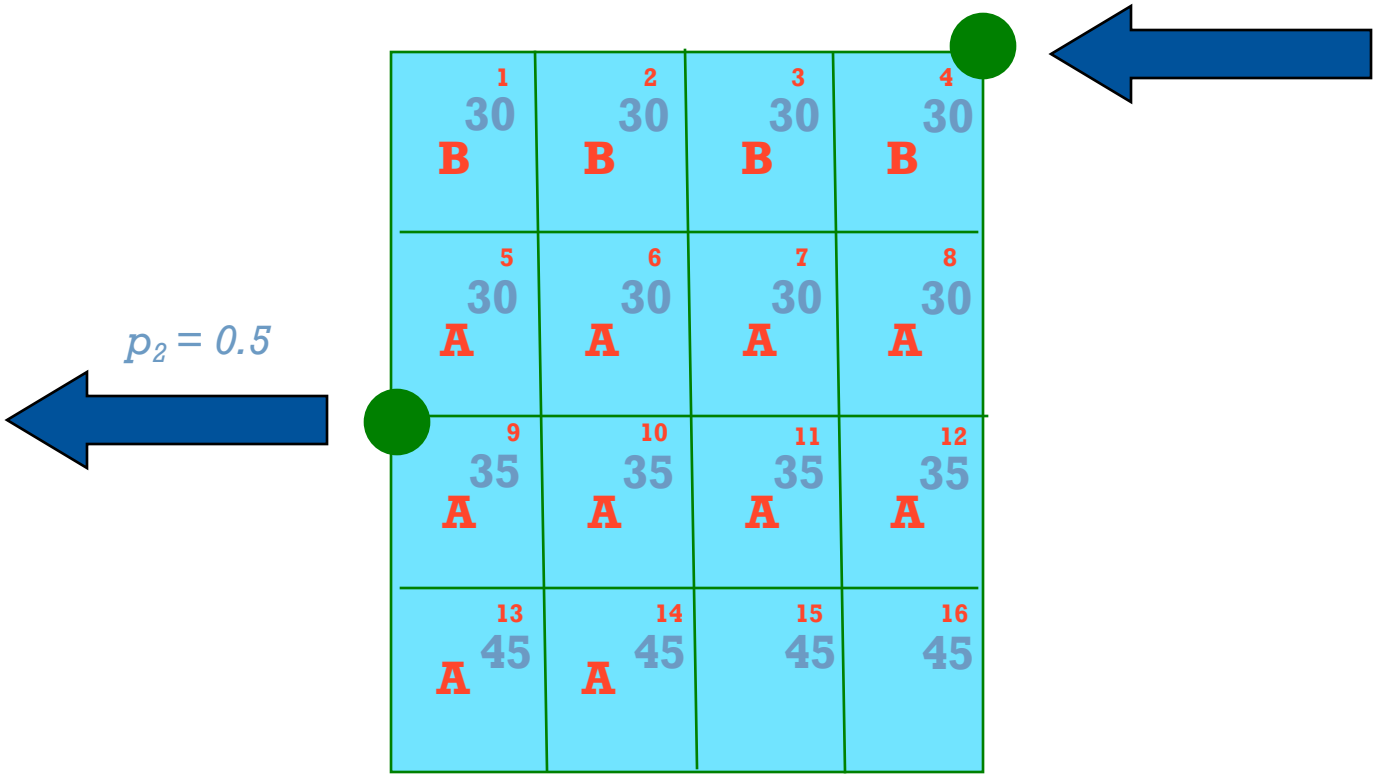
$T_B=80$
$S_B=4$
$T_B/S_B=20$

Consider Product B first and Product A second.

$T_B/S_B > T_A/S_A$

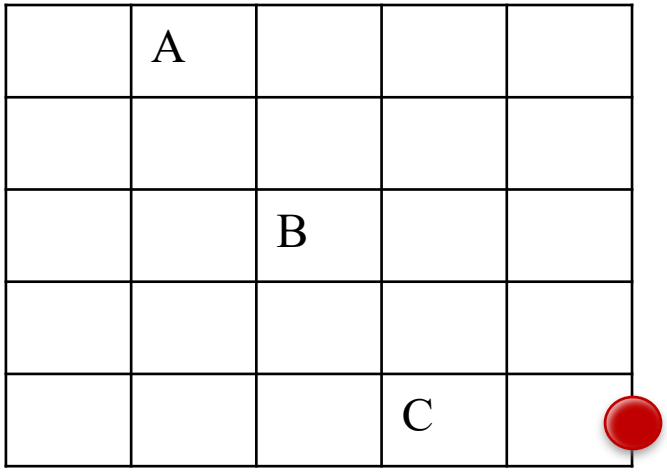
$p_1 = 0.5$

$p_2 = 0.5$



In-class Activity

The Heavener Football Center stores helmets, jerseys, and pads. The facility's storage is represented below, where each grid represents a storage location, and the only I/O point is shown in red. Assume 13 storage locations are needed for helmets, 8 for pads, and 4 for jerseys. All items are retrieved and stored in equal quantities. What items should be stored in A, B, and C?



In-class Activity

Answer

A=Helmets
B=Pads
C=Jerseys